

Markscheme

Mathematics: analysis and approaches
Practice question bank

102. (a)

It can be proven by contradiction that if n^2 is even, then n is even, for every integer n (assume n is odd, then show n^2 must also be odd, a contradiction).

Using this result, find the value of the even integer n satisfying $n^2 = 196$.

$$n = \pm\sqrt{196} = \pm 14 \quad (M1)$$

since n must be even, and both ± 14 are even, take the positive value **R1**

$$n = 14 \quad \text{A1}$$

[4 marks]